scientist Robert Axelrod (working partly in collaboration with W. D. Hamilton, whose name has cropped up on so many pages of this book) has taken the idea of reciprocal altruism on in exciting new directions. It was Axelrod who coined the technical meaning of the word 'nice' to which I alluded in my opening paragraph.

Axelrod, like many political scientists, economists, mathematicians, and psychologists, was fascinated by a simple gambling game called Prisoner's Dilemma. It is so simple that I have known clever men misunderstand it completely, thinking that there must be more to it! But its simplicity is deceptive. Whole shelves in libraries are devoted to the ramifications of this beguiling game. Many influential people think it holds the key to strategic defence planning, and that we should study it to prevent a third world war. As a biologist, I agree with Axelrod and Hamilton that many wild animals and plants are engaged in ceaseless games of Prisoner's Dilemma, played out in evolutionary time.

In its original, human, version, here is how the game is played. There is a 'banker', who adjudicates and pays out winnings to the two players. Suppose that I am playing against you (though, as we shall see, 'against' is precisely what we don't have to be). There are only two cards in each of our hands, labelled COOPERATE and DEFECT. To play, we each choose one of our cards and lay it face down on the table. Face down so that neither of us can be influenced by the other's move: in effect, we move simultaneously. We now wait in suspense for the banker to turn the cards over. The suspense is because our winnings depend not just on which card we have played (which we each know), but on the other player's card too (which we don't know until the banker reveals it).

Since there are 2 × 2 cards, there are four possible outcomes. For each outcome, our winnings are as follows (quoted in dollars in deference to the North American origins of the game):

Outcome I: We have both played COOPERATE. The banker pays each of us $300. This respectable sum is called the reward for mutual cooperation.

Outcome II: We have both played DEFECT. The banker fines each of us $10. This is called the punishment for mutual defection.

Outcome III: You have played COOPERATE; I have played DEFECT. The banker pays me $500 (the temptation to defect) and fines you (the sucker) $100.